

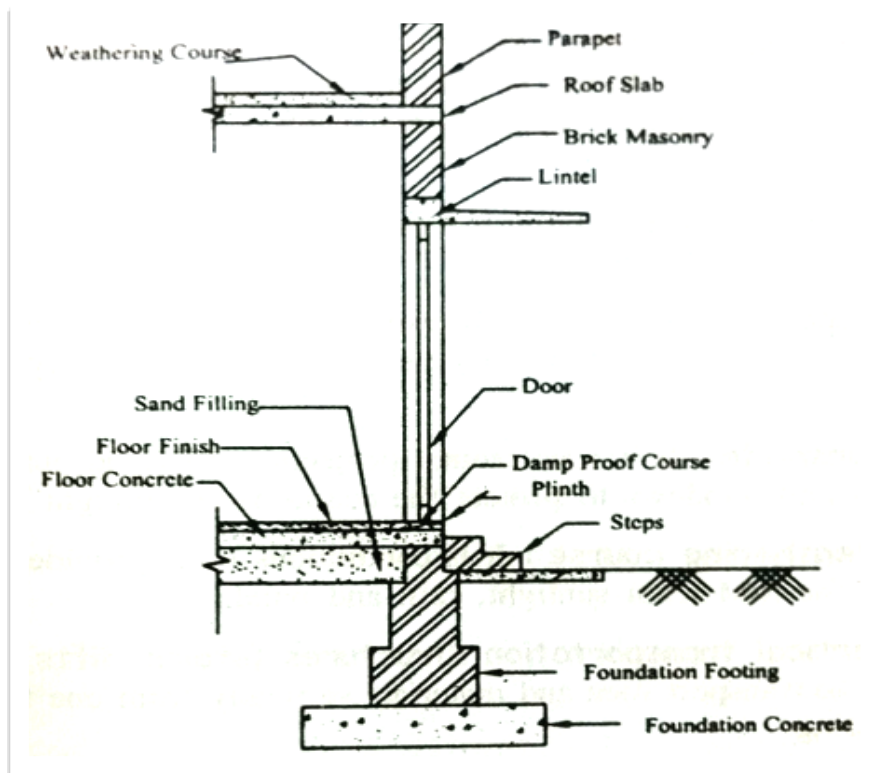
UNIT 2

BUILDINGS AND THEIR COMPONENTS

BASIC COMPONENTS OF THE BUILDING

Sub-structure – The part of the structure which is constructed below ground level (foundation)

Super-structure - The part of the structure which is constructed above ground level (wall, roof, floor, wall, lintel, column, beam, roof, stairs, sunshade)



Components of a Building

FUNCTIONS OF THE VARIOUS COMPONENTS OF A BUILDING

1. **Foundation:** Foundation is the most important part of the building. It is the lower most part of the building. It transfers the load of the building to the ground below it.
2. **Plinth:** The portion of the wall between the ground level and the ground floor level is called plinth. It is usually of stone masonry.
3. **Walls & Columns:** The function of walls and columns is to transfer the load of the structure vertically downwards to transfer it to foundation. Apart from this wall performs the function of dividing building area into different compartments and provides privacy.

4. **Beam:** Beam is a horizontal member that transfers the load of a roof or floor slab to the wall or columns that support it. It can be made of wood or stone for small spans and RCC or steel for longer spans.
5. **Lintels:** Lintels are the RCC or stone beams provided over the door and window openings to transfer the load transversely so as to see that door or window frame is not stressed unduly.
6. **Doors:** The function of a door is to give access to different rooms in the building and to deny the access whenever necessary. The size of the door should be of such dimension as will facilitate the movement of the largest object likely to use the door.
7. **Windows:** Windows are provided to get light and ventilation in the building. In hot and humid regions, the window area should be 15 to 20 percent of the floor area.
8. **Floors:** Floor divides the building into different levels or storeys. It also provides the horizontal plane surface where the various functions of a building occur.
9. **Roof:** It is the flat portion used to cover entire building and protect the inmates from weathering agencies such as rain, sun, etc.
10. **Steps:** Steps give convenient access from ground level to ground floor level.
11. **Weathering course:** It is the continuous layer provided over the roof slab to protect the roof from natural agencies like wind, rain, sun.

SOILS

Soil is the un-aggregated deposits of minerals and organic particles covering large portion of the earth's crust. It is produced by the weathering of the soil rock.

TYPES OF SOILS

On the basis of the origin of constitute of soils can be divided into two large groups

- i. Residual soil
- ii. Transported soil

Residual soil: Those which remain at the place of their formation as a result of weathering of parent rock.

Transported soil: Any soil that has been transported from its place of origin by wind, water, ice or some other agency and has been re-deposited is called as transported soil. This is further classified based on the weathering action as follows.

- Deposited from suspension in running water – Alluvial soil
- Deposited in quiet lakes- Lacustrine soil
- Deposited from suspension in sea water – Marine soil
- Deposited by wind – Aeolin soil

- Deposited through ice – Glacial ice
- Deposited through gravitational force – Colluvial soil

CLASSIFICATION OF AGGREGATES BASED ON GRAIN SIZE

Coarse aggregates have particle size > 4.75 mm.

Ordinary stones and pebbles are example for coarse aggregates.

Fine aggregates have particle size < 4.75 mm.

Common sand and all natural soils are example for fine aggregates.

IS CLASSIFICATION OF AGGREGATE

Boulder	Cobble	Gravel		Sand		Silt	Clay
>300 mm	80-300 mm	Coarse	Fine	Coarse	Fine	75-2 micron	<2 micron
		20-80 mm	4.75-20 mm	4.75mm -425 micron	425-75 micron		

BEARING CAPACITY OF SOIL

The load carrying capacity of soils is called as their bearing capacity. In another words the property of soils to take the loads without failing is known as bearing capacity of soils.

ULTIMATE BEARING CAPACITY

The maximum load the soil can carry without failure is known as its ultimate bearing capacity.

SAFE BEARING CAPACITY OF SOIL

The maximum load the soil can carry safely without the risk of failure is known as its safe bearing capacity.

$$\text{safe bearing capacity} = \frac{\text{ultimate bearing capacity}}{\text{Factor of safety}}$$

FACTORS THAT AFFECT THE BEARING CAPACITY OF SOIL

- Magnitude and type of load on the soil
- Type, shape, size and depth of footing
- Depth of water table below foundation
- Permeability and moisture content of soil
- Cohesion and angle of internal friction of the grains of the soil
- Density and compactness of soil

Methods of improving bearing capacity of soils

(i) Surface compaction

Reduction in air voids present in the soil mass increase the bearing capacity of soil. This is done by compacting the soil. It may be effected by ramming and/or by driving piles.

(ii) Increase the depth of foundation

This method can be adopted only when the increase in the cost of construction by increasing the depth of footing is not much. It cannot be adopted where the ground water level is high.

(iii) Draining the sub-soil water

The water present in the soil is drained by laying open jointed drains in trenches. Reduction in water content increases the bearing capacity of the soil.

(iv) Grouting and injection

Cement grout is injected into the soil so that the cracks and fissures are filled. Chemicals like silicates are injected into the soil mass, so that the soil becomes a compact unit.

(v) Soil reinforcements

Weak soils are reinforced by high strength thin horizontal member. Materials like timber, straw, steel, concrete, glass fibre, etc., are used as reinforcing materials.

(vi) Confining the soil mass

The bearing capacity of soil can be improved by restricting its movement. This can be done by providing sheet pile enclosure.

FOUNDATION

Foundation is part of the sub-structure of a building, which is below the ground level and in direct contact with the ground. It safely transmits and spreads the load from the superstructure to the soil below it.

TYPES OF FOUNDATION:

SHALLOW FOUNDATIONS

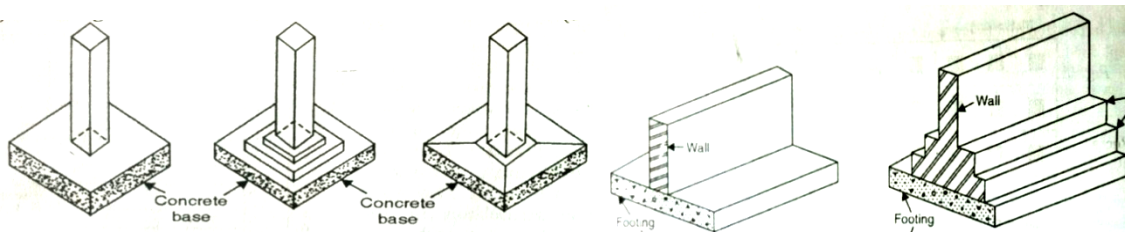
Shallow foundations are those foundations whose width is more than their depth.

FOOTING

Footing may be provided for a column or wall or combinations of these two. It is the basic form of foundation.

TYPES OF FOOTINGS

1. **Isolated footing** : An isolated footing is provided under column to transfer the load safely to the soil bed. If the column is loaded lightly, a spread is given under the base of the column. This spread is known as footing.

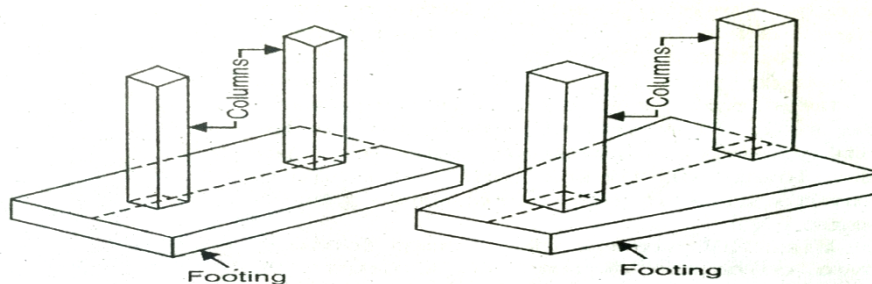


For heavy loaded column- **stepped footing**.

For gradually increasing the width towards the bottom -**sloped footing**.

If the footing is provided throughout the length of the wall in the case of load bearing walls, then it is called **wall footing**. Wall footings can be either simple or stepped footings.

2. **Combined footing**: If a footing is constructed for two or more columns is called as combined footing. Generally, the shape of the footings is rectangular or trapezoidal.



Combined footing

3. **Strap footing**: If two or more footing of individual columns is connected by a beam, it is called Strap footing.

4. Continuous footing

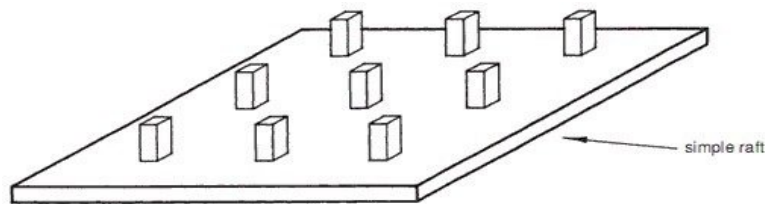
In this type of footing, a single continuous R.C. slab is provided as foundation for three or more columns in a row. This type of footing is more suitable to prevent the differential settlement in the structure and for the safety against earthquake.



Continuous footing

RAFT OR MAT FOUNDATION

When the loads are heavy and the space for foundation base is less or bearing capacity of soil is low, Raft or Mat foundations are provided. They are like an inverted slab-beam – column system. It is useful for Public building, office building, school building etc.



Mat foundation

DEEP FOUNDATIONS

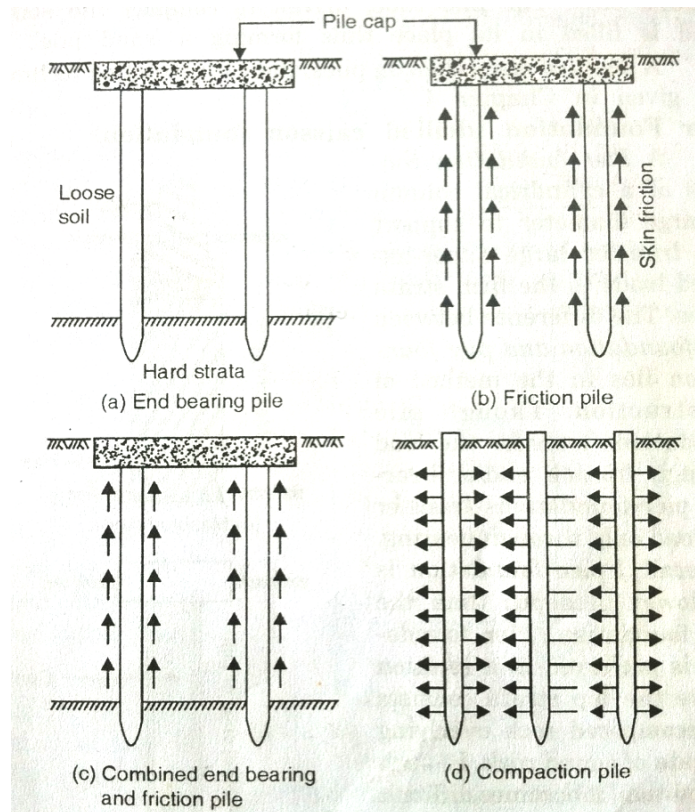
Deep foundations are those foundations whose depth is more than their width. They are constructed when the bearing capacity of soils are too less to take load from shallow foundations or the loads from the structure are very high.

Under the following circumstances, the piles may be used as foundation:

- The live load and dead load coming from the structure is very heavy and the distribution of load on soil is uneven.
- The subsoil water level is likely to raise or fall appreciably
- The piles are preferable in marine structure.

1. Pile foundation:

Piles are slender cylindrical members used as foundations to transfer loads from substructure to the soil below it. A group of piles are connected at ground level by a pile cap and the substructure is constructed above the pile cap.



Types of Pile foundation

End bearing piles: This pile penetrates through the soft soil and their bottom rests on a hard bed. Thus, these piles act as columns or piers. The soft ground through which the piles pass also gives some lateral support to the piles.

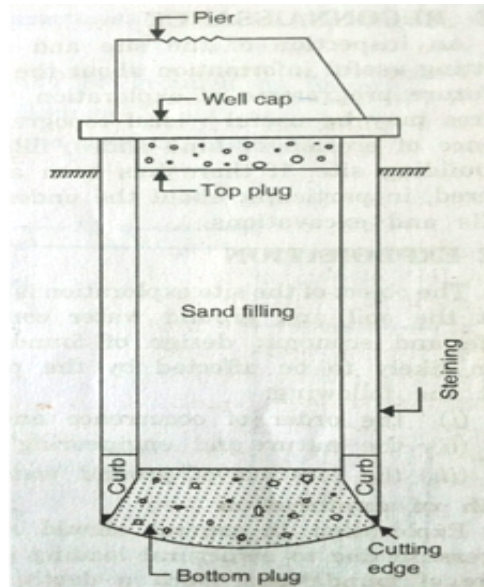
Friction piles: When loose soil extends to a great depth, piles are driven up to such a depth and frictional resistance developed at the sides of the piles equals the load coming on the piles.

Compaction piles: Compaction piles are used to compact loose granular soils, thus increasing their bearing capacity.

Under ream piles: These are the piles with one or more bulbs in its vertical shaft. These bulbs are known as under ream and they increase the bearing capacity of the pile considerably and also resist the uplift against horizontal forces (like wind).

WELL/CAISSON FOUNDATION:

This is generally used for foundation under water such as for ducts, bridges, etc. It will not be a solid structure like that of the pier but will be hollow inside, resembling a well. The load is transferred through the wall around called **staining**. The well is constructed and brought to the site. Then it is gradually driven down by digging the soil from inside. The bottom is plugged with concrete and the hollow portion is filled with sand. The whole well is then covered with a well cap above which the super structure will be constructed.



Well foundation

FUNCTIONS OF FOUNDATIONS

- (a) Distribute the load from the structure to soil evenly and safely.
- (b) To anchor the building to the ground so that under lateral loads building will not move.
- (c) It prevents the building from overturning due to lateral forces.
- (d) It gives level surface for the construction of super structure.
- (e) To protect the building against scouring and undermining by flood water, drainage water, etc.,

FAILURE OF FOUNDATION

(a) Unequal settlement of supporting soil: Unequal settlement of supporting soil can result from

- (i) Non-uniform pressure distribution,
- (ii) Non-uniform sub-soil condition,
- (iii) Overstressing due to adjacent building

Remedial measures:

- (i) Placing the foundation on hard soil or rock,
- (ii) Limiting pressure on soil
- (iii) Avoid eccentric loading

(b) Sub-soil moisture movements: Seasonal expansion and shrinkage of foundation of soil cause footing swell and crack.

Remedial measures: the foundation has to be taken deeper and a gap may be provided below the floor allowing for expansion of soil

(c) Compression of mortar joints Excessive loading of masonry before the mortar sets causes unequal settlements of masonry.

Remedial measures: using thin and adequate strength of mortar joints, proper curing of joints.

(d) Movement of Sub-soil: Foundation placed on slope and subjected to movement of water may laterally causing excessive settlement and reduction of shearing resistance.

Remedial measures: Providing lateral support by sheet pile wall or proper drainage system.

(e) Atmospheric actions: The causes of failure in footing due to atmospheric are removal of soil due to scouring action of rain and wind.

Remedial measures: Providing adequate drainage arrangements and sun shade in building,

(f) Weathering Action: The causes of failure in footing due weathering actions is roots of trees and shrubs may consume more underground moisture

Remedial measures: Providing some moisture barrier and be avoided by growing trees and shrubs away from the building.

MASONRY

Masonry is the part of building which is constructed using the building units or blocks such as bricks or stones and are bonded together with binding material like mortar (lime or cement mortar).

TYPES OF MASONRY

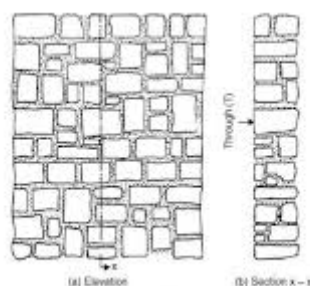
- Brick masonry
- Stone masonry

TYPES OF STONE MASONRY

RUBBLE MASONRY

In this type of masonry, stones of irregular sizes and shapes are used. The stones, as obtained from quarry, are taken in use in the same form or they are broken and shaped in suitable sizes by means of hammer as the work proceeds.

1. **Random rubble masonry:** In this type of masonry, the stones used are of widely different sizes. This is the roughest and cheapest form of stone masonry.



Un coursed rubble masonry



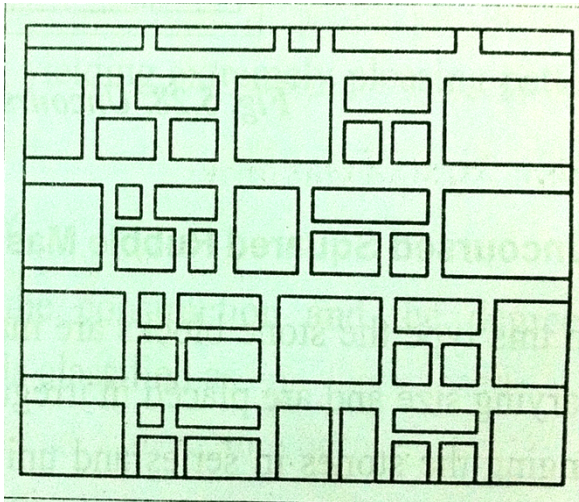
Coursed rubble masonry

- In this coursed random rubble masonry, the masonry work is carried out in courses such that the stones are of equal heights.
- In un-coursed random rubble masonry, the masonry courses are not maintained regularly. The larger stones are laid first and the spaces between them then filled up by means of spalls or snecks.

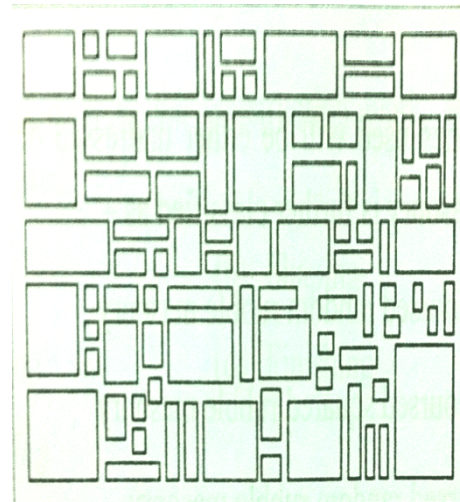
2. Square rubble masonry

In this type of masonry stones having straight bed and sides are used. The stones are usually squared and brought to a hammer dressed or straight cut finish.

- In the coursed square rubble masonry, the work is carried out in courses of varying depth.
- In the un-coursed square rubble masonry, the different sizes of stones having straight edges and sides are arranged on face in several irregular patterns.



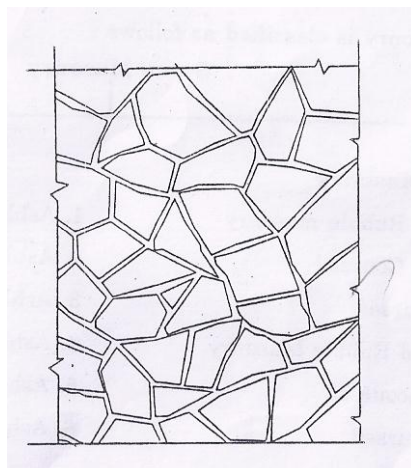
Coursed square rubble masonry



Un Coursed square rubble masonry

3. Polygonal rubble masonry

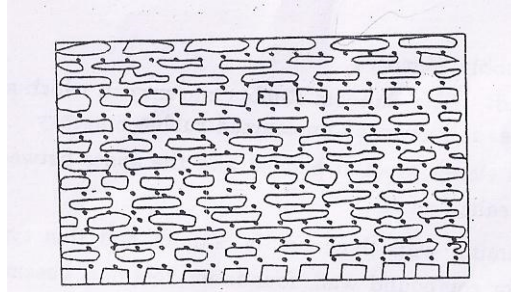
In this type of rubble masonry, the stones are hammer dressed. The stones used for face work are dressed in an irregular polygonal shape. Thus the face joints are seen running in an irregular fashion in all directions.



Polygonal rubble masonry

4. Flint rubble masonry

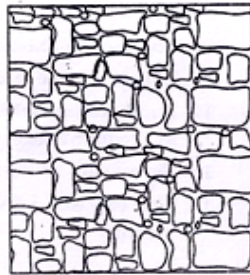
In this type of rubble masonry, stones used are flints or cobbles. There are irregularly shaped nodules of silica. The stones are extremely hard. But they are brittle and therefore they break easily. The face arrangements may either coursed or un-coursed.



Flint rubble masonry

5. Dry rubble masonry

In this type of masonry, mortar is not used in the joints. This type of constructions is the cheapest and requires more skill in constructions. This may be used for non-load bearing walls such as compound wall etc.



Dry rubble masonry

Advantages of rubble masonry

- Rubble masonry is cheaper than ashlar masonry as cutting and dressing of stones are not required and hence less labour and cost.
- Rubble masonry is easier to construct than ashlar masonry
- Rubble masonry requires less skilled labour than ashlar masonry
- Rubble masonry is faster to construct than ashlar masonry

Disadvantages of rubble masonry

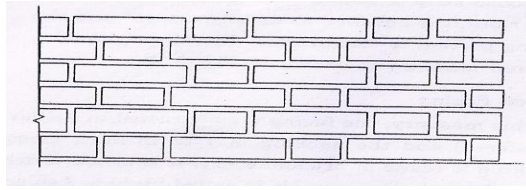
- Rubble masonry gives unpleasant look than ashlar masonry
- Rubble masonry requires more stones than ashlar masonry
- Rubble masonry cannot be used for facing works like compound walls and outer walls.
- They are preferred for foundations and unimportant works.

ASHLAR MASONRY

In this type of constructions, no irregular stones are used. The stones used in this masonry are rectangular blocks and are all dressed finely with chisel.

1. Ashlar fine masonry

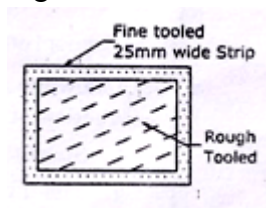
In this type of ashlar masonry, each stone is cut to uniform size and shape with all sides rectangular, so that the stone gives perfectly horizontal and vertical joints with adjoining stone. This type of ashlar masonry is very costly.



Ashlar fine masonry

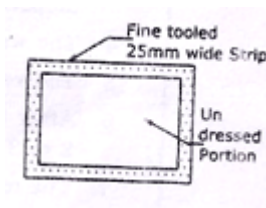
2. Ashlar rough tooled masonry

In this type of ashlar masonry, the beds and sides are finely chisel-dressed. But the face is made rough by means of tools. A strip, about 25mm wide and made by means of a chisel is provided around the perimeter of the rough dressed face of each stone.



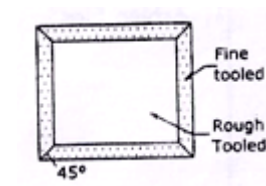
3. Ashlar rock or quarry faced masonry

In this type of ashlar masonry, a strip about 25mm wide and made by means of chisel is provided around the perimeter of every stone as in case of rough- tooled ashlar masonry. But the remaining portion of the face is left in the same form as received from quarry.



4. Ashlar chamfered masonry

In this type of ashlar masonry, the strip is provided as above. But is chamfered or bevelled at an angle of 45 degrees by means of chisel for a depth of about 25mm.



Ashlar chamfered masonry

5. Ashlar block-in-coarse masonry: This is combination of rubble masonry and ashlar masonry. In this type of masonry, the face work is provided with rough tooled or hammer dressed stones and backing of the wall may be made in rubble masonry.

❖ Advantage of ashlar masonry

- Rubble masonry gives unpleasant look than ashlar masonry
- Rubble masonry requires more than ashlar masonry
- Rubble masonry cannot be used for facing works like compound walls and outer walls. They are preferred for foundations and unimportant works.

❖ Disadvantages of ashlar masonry

- Rubble masonry is cheaper than ashlar masonry as cutting and dressing of stones are not required and hence less labour and cost.
- Rubble masonry is easier to construct than ashlar masonry
- Rubble masonry requires less skilled labour than ashlar masonry
- Rubble masonry is faster to construct than ashlar masonry

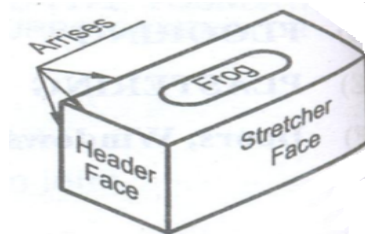
POINTS TO BE CONSIDERED WHILE CONSTRUCTING THE STONE MASONRY

The following points should be kept in mind in supervising stone masonry work:

- Hard and durable stones, free from defects like flaws, cavities veins etc. should be used.
- Dressing of the stones should be as per the requirement.
- Stones should be properly wetted before they are used so as to avoid sucking of water from mortar.
- Stones should be laid on their natural bed.
- The heart of masonry should be filled with stone chips and mortars. To thick mortar joints should be avoided.
- Verticality of the wall should be frequently checked with plumb-bob.
- Mortars with correct proportion of sand and cement should be used.
- Continuous vertical joints should be avoided.
- Under the beams, trusses, sill's etc large flat stones should be used.
- Before continuing work, the masonry built on previous day should be well cleaned and freed from loose particles.
- Curing should be done properly for 2 to 3 weeks.

BRICK MASONRY:

Brick masonry is built with bricks bonded together with mortar.



COURSE AND BED

- Course is a horizontal layer of masonry unit
- Bed is the lower surface of a masonry unit in each course

LAP AND BED JOINT

- Lap is the horizontal distance between the vertical joints of successive brick courses
- A joint formed by bed (or lower surface) of a course and the upper surface of the course below it is called bed joint

HEADER COURSE AND STRETCHER COURSE

- A course of brick showing only the headers on the exposed surface the wall is called header course or heading course
- A course of brick showing only the stretchers on the exposed surface the wall is called stretcher course or stretching course

CLOSER

Closer is the portion of a brick cut along its length, such that its width is shortened. Thus closer is a header of smaller width

BAT: Bat is a part of a brick cut across its length, such that its length is shortened. Thus bat is a stretcher of smaller length

KING CLOSER AND QUEEN CLOSER

King closer is the portion of a brick obtained by cutting away the triangular portion between the centre of one shorter end and the centre of its longer adjacent end.

Queen closer is the portion of a brick cut into two equal halves along its length. Thus the width of the header is reduced

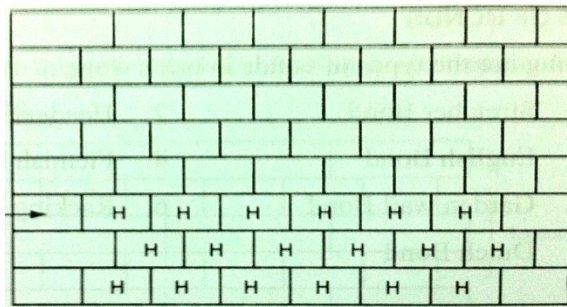
TYPES OF BONDS IN BRICK MASONRY

1. Stretcher bond: It has the stretchers arranged to face the side of wall. It can be used for even $\frac{1}{2}$ brick wall.



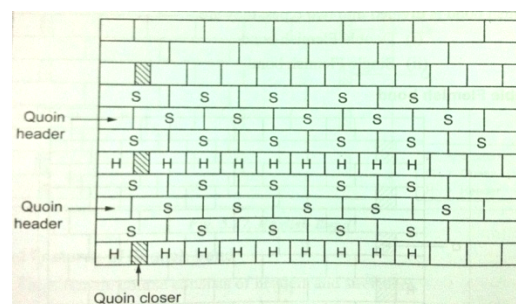
Stretcher bond

2. Header bond: It has the headers arranged to face the side of wall. It requires at least 1 brick thick wall.



Header bond

3. English bond: Alternate courses consist of headers and stretchers.



English bond

4. Single Flemish bond: Headers and stretchers are arranged alternatively in same course along facing.

H	3/4	H	S	S	S	
S		S	S	H	S	S
H	3/4	H	S	S	S	
S		S	S	H	S	S
H	3/4	H	S	S	S	
S		S	S	H	S	S
H	3/4	H	S	S	S	

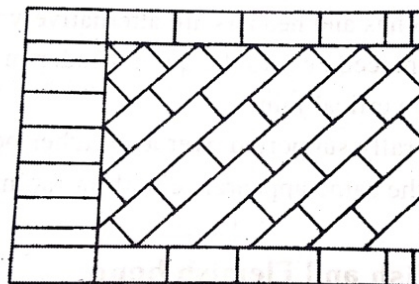
Single Flemish bond

5. Double Flemish bond: Headers and stretchers are arranged alternatively in same course along both sides of wall.

6. Raking bond: In this type of bond, the bonding bricks are kept at an inclination to the direction of the wall. The raking course is generally provided between the two stretcher courses. This bond is used in thick walls.

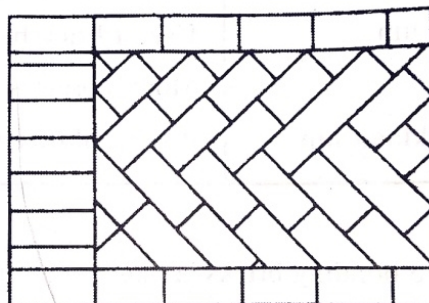
Types: Diagonal bond and Herring bone bond

Diagonal bond: in this bond the bricks are laid diagonally. The angle of inclination is so selected that there is minimum breaking of the bricks. The triangular pieces of bricks are required near the sides.



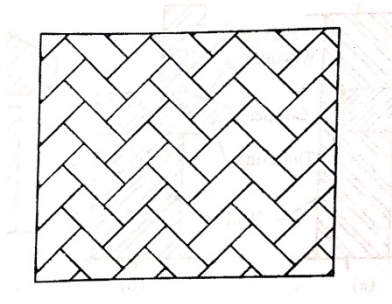
Diagonal bond

Herring bone bond: In this bond, the bricks are laid at an angle of 45° from the centre in both the directions.



Herring bone bond

7. **Zigzag bond:** This bond is similar to herring bone bond, except that the bricks are laid in zigzag fashion as in figure. This bond is commonly used for making ornamental panels in the brick flooring.



Zigzag bond

COMPARISON OF ENGLISH AND FLEMISH BOND IN MASONRY

English bond	Flemish bond
Alternates courses consists of headers and stretchers	Headers and stretchers are arranged alternatively in same course
For walls thicker than 1 ½ bricks its is more stronger	For thicker walls its is comparatively weaker
Appearance is not pleasing	Appearance is pleasing
Preferred for walls with plastered faces	Preferred for walls with exposed faces
Skilled labour is not required	Skilled and experience labour is required
Mortar requirement is less	Mortar requirement is more as more brick bats are used
It is the most commonly used bond	It is not commonly used bond

POINTS TO BE CONSIDERED WHILE CONSTRUCTING THE BRICK MASONRY

The following points should be observed in the construction of brick masonry:

- Before using the bricks in masonry, they should be soaked in water for 2 hours so that bricks do not absorb water from the mortar.
- Bricks should be laid with the frog pointing upward.
- Construction of brick wall should start from the end or corner.
- Brick courses should be perfectly horizontal.
- Verticality of the wall should be ensured by frequently checking with plumb-bob.
- Mortar used should be as per specification.
- Whenever work is stopped brick masonry should be left with toothed end.
- Use of brick bats should be avoided.
- Walls should be raised uniformly. In no case difference between adjoining walls is more than 1 m. In a day no wall should be raised by more than 1.5 m.
- Brick masonry should be regularly cured for 2 weeks.

COMPARISON OF BRICK MASONRY AND STONE MASONRY

Sl.No.	Criteria	Brick Masonry	Stone Masonry
1	Available	Manufactured using clay and bricks are available around cities	available are to be brought from quarries which are located only at few places
2	Shape and size	uniform	Non uniform
3	Mortar use	use all types of mortar even mud mortar can be used.	More quality of mortar required.
4	Wall Thickness	Thinner walls can be constructed	Difficult to construct walls less than 23cm
5	Form openings	easy to form openings for doors and windows	Dressing of stones are required to achieve this
6	Dead load	less	Heavy
7	cost	Less	High
8	Labour	Semi-skilled labour is needed	Skilled labour is necessary
9	Compressive strength	Less	Very high
10	Durability	less	High
11	Plastering	Needed and plastered surface needs colour washing	Not needed
12	Maintenance	More	Less
13	Moisture absorption	Absorbs water and there are possibility of dampness.	There is no such problem
14	Appearance	Elegant appearance .	Gives massive appearance
15	Lifting device	Not needed	Needed

FLOOR

Floor is a horizontal member of a building. It divides the building into different levels or storeys, for creating accommodation within a restricted space. It also provides the horizontal plane surface where the various functions of a building occur.

Example: in a residential building people live or reside on floors.

COMPONENTS OF THE FLOOR

Floor slab, floor base or floor bed and floor finish are the components of floors.

REQUIREMENTS OF FLOORS

- Economical in construction
- Ease to maintenance
- Smooth and good appearance
- Durability and hardness
- Fire resistance and damp resistance
- Thermal insulation and sound insulation

FUNCTIONS OF FLOOR

- It divides the building into different levels or storeys.
- It also provides the horizontal plane surface where the various functions of a building occur.

Example:

- In a residential building people live or reside on floors and in storage building things are stored on floor
- A floor of upper level also acts as roof of lower level
- The ground floor provided damp proofing to the building above it.

TYPES OF FLOORS

Based on the material of construction, types of floors are stone floor, brick floor, timber floor, concrete floor, rubber floor, glass floor and plastic or PVC floor.

Based on floor finish, the types of floor are mud flooring, cement flooring, marble flooring, mosaic flooring and tiled flooring.

1. Mud flooring

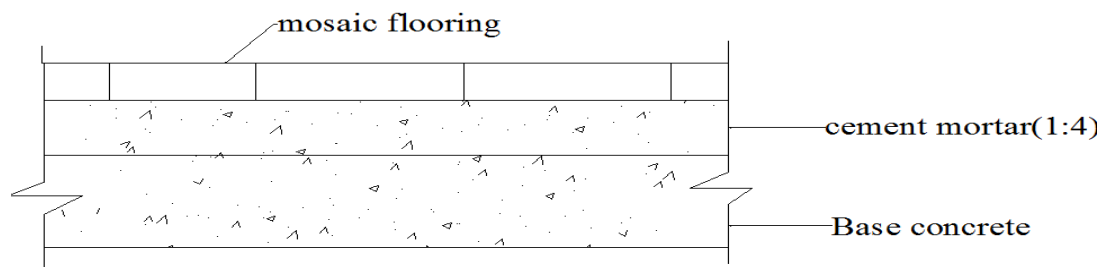
This floor is used in low cost housing, especially in villages. The floor bed is well prepared and a 250mm thick layer of selected moist earth is evenly spread out and rammed to get a consolidated thickness of 150mm, without using water. The floor needs a thin wash of cow dung at least once a week. This floor is cheap, hard and easy in construction and maintenance.

2. Brick flooring

This is a cheap floor construction; it is commonly used in godowns and factories. After preparing the sub grade, 10 to 15mm thick lean concrete or lime concrete is laid over it. This forms the base course and over this good quality bricks are laid flat or on edges in desired pattern on 12mm thick mortar bed. Joints are rendered flush and finished with cement mortar. This floor is cheap, non-slippery, durable, sufficiently hard and easily repairable.

3. Mosaic flooring.

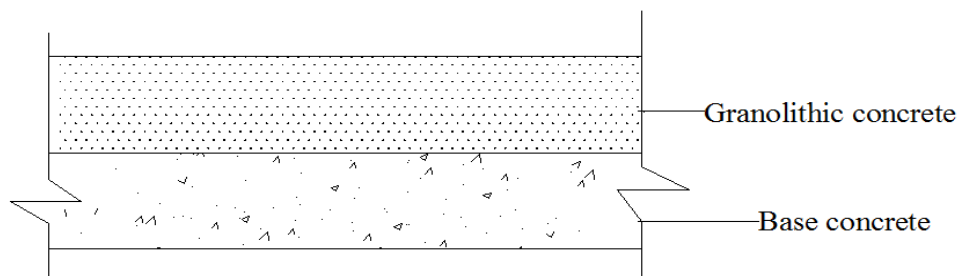
It consists of slabs with a coat of small pieces of broken tiles of china glazed or a marble arranged in different patterns, set in lime or cement mortar. The base course is concrete flooring and on it 30 to 40mm cement mortar layer is provided. On this mortar layer slabs of broken pieces tile of china glazed or marble are laid. Coloured cement is sprinkled and pressed in joints. After 24 hours of drying the top is rubbed with corborundum stone to get smooth and polished surface. This type of floor is used in homes, public buildings, etc.



Mosaic flooring

3. Granolithic flooring

This floor is similar to cement concrete flooring. Base course is laid over well compacted soil. Its thickness is usually 75mm to 100mm. It consists of lean cement concrete mix (1:4:8) or lime concrete containing 40% of 1:2 lime mortar and 60% of coarse aggregate of 40mm size. After base course is hardened wearing coat 40mm is laid with rich concrete mix in which sand is replaced with crushed granite powder.



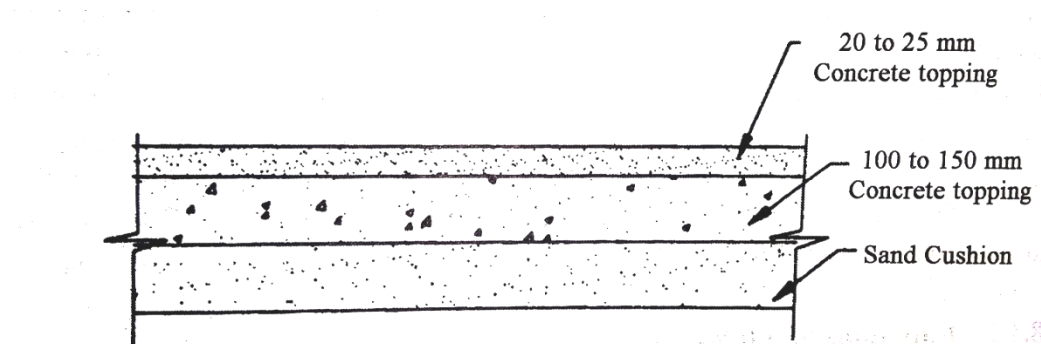
Granolithic flooring

4. Cement concrete flooring

It is modestly cheap and durable floor and hence commonly used in residential, commercial and industrial buildings. It consists of two courses-base courses and wearing coat. Base course is laid over well compacted soil. Its thickness is usually 75mm to 100mm. It consists of lean cement concrete mix (1:4:8) or lime concrete containing 40% of 1:2 lime mortar and 60% of coarse aggregate of 40mm size. After base course is hardened wearing coat of 40mm is laid. It consists of panels of 1m x 1m, 2m x 2m or 1m x 2m.

Alternate panels are laid with 1:2:4 concrete using wooden, glass or asbestos strip separators of 1.5mm to 2.0mm thickness. To get good bond between base coarse and wearing coat cement slurry wash is given before laying wearing coat panels. After 3 to 4 days of laying of one set of panel, another alternate panel are laid. Top of these are finished by tamping the surface with wooden floats and tapping with trowels, till cement slurry appears on top. It

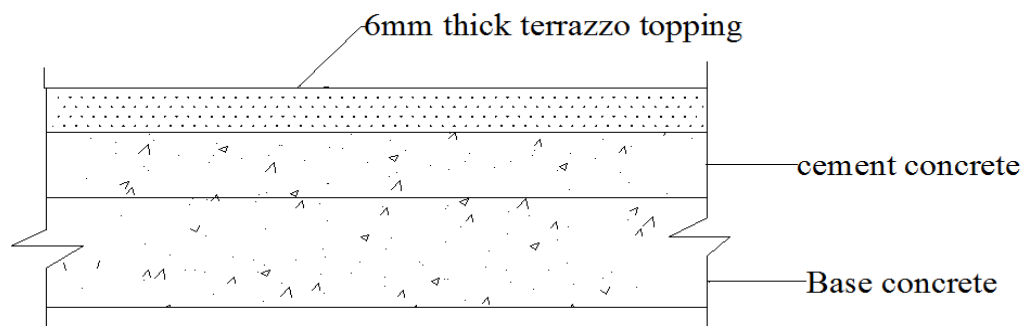
needs curing for 7 to 14 days. To get good appearance many times red-oxide finishing coat is provided. It is cheap and durable, fire resistant and easy to maintain.



Cement concrete flooring

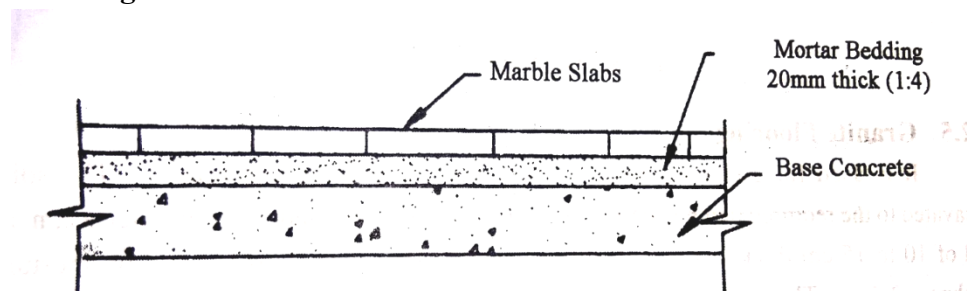
5. Terrazzo flooring

This floor is similar to cement concrete flooring. Base course is laid over well compacted soil. Its thickness is usually 75mm to 100mm. It consists of lean cement concrete mix (1:4:8) or lime concrete containing 40% of 1:2 lime mortar and 60% of coarse aggregate of 40mm size. After base course is hardened wearing coat of 40mm is laid with concrete of (1:2:4) and 6mm of terrazzo topping. Terrazzo is prepared by mixing concrete with white cement and marble chips of various colours.



Terrazzo flooring

6. Marble flooring



Marble flooring

It is superior type of flooring used in kitchens and bathrooms of residential buildings, hospitals, etc, where extra cleanliness is an essential requirement. It may be laid in different sizes, usually in rectangular or square shapes. The base concrete is prepared in the same manner that for concrete flooring. Over the base concrete, 20mm thick cement mortar of 1:4 mix is spread under the area of each individual slab. Then the marble slab is laid over it,

gently pressed with wooden mallet and levelled. The marble slab is then lifted up and fresh mortar is added to the hollows of the bedding mortar. After hardening slightly, the cement slurry is spread over it. The floor is cured for about a week.

ROOFS

It is the upper most portion of the building which protects the building from rain, wind and sun. Various types of roofs used may be divided broadly into three types:

1. Flat roofs
2. Pitched roofs
3. Shells and folded plates.

Flat roofs are used in plains where rainfall is less and climate is moderate. Pitched roofs are preferred wherever rainfall is more. Shells and folded plate roofs are used to cover large column free areas required for auditoriums, factories etc.

1. Flat Roofs: These roofs are nearly flat. However slight slope (not more than 10°) is given to drain out the rain water. All types of upper storey floors can serve as flat roofs.

The **advantages** of flat roofs are:

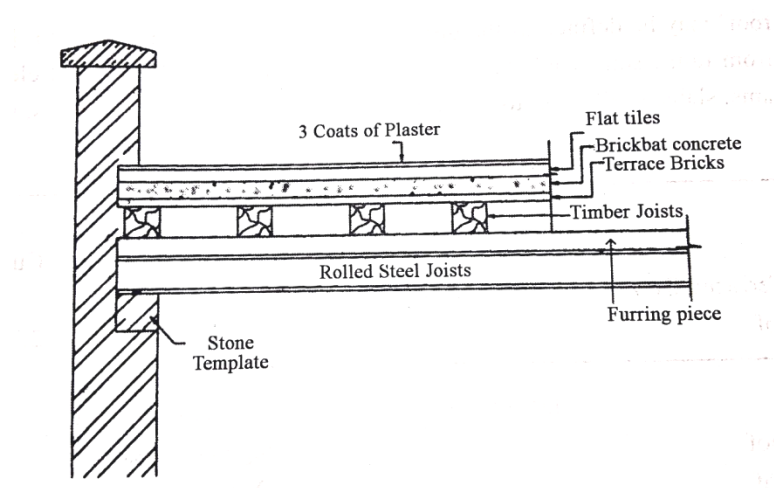
- (a) The roof can be used as a terrace for playing and celebrating functions.
- (b) At any latter stage the roof can be converted as a floor by adding another storey.
- (c) Over-head water tanks and other services can be located easily.
- (d) They can be made fire proof easily compared to pitched roof.

The **disadvantages** of flat roofs are:

- (a) They cannot cover large column free areas.
- (b) Leakage problem may occur at latter date also due to development of cracks. Once leakage problem starts, it needs costly treatments.
- (c) The dead weight of flat roofs is more.
- (d) In places of snow fall flat roofs are to be avoided to reduce snow load.
- (e) The initial cost of construction is more.

Types of Flat Roofs:

1. Madras Terrace roof

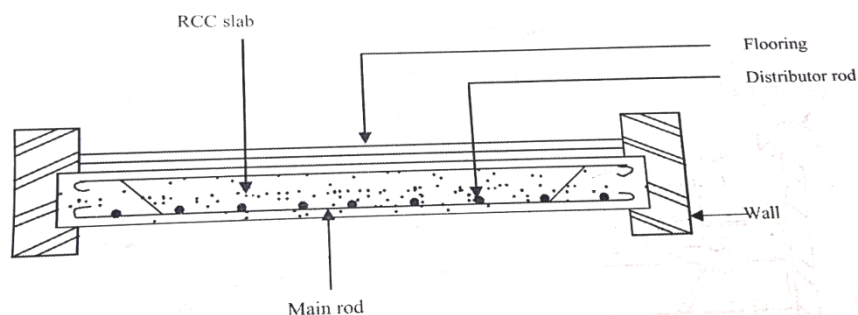


- Wooden joist, usually a teak wood are placed over supporting walls or steel joists.
- A course of well burnt and specially prepared bricks is laid in lime mortar keeping the bricks on their edges diagonally across the joists.
- After the brick course complete set, 10cm thick layer of brick bat concrete is laid over it. It is frequently wetted with water.
- After the concrete has set, flat tiles are laid over it in lime 1: 1. The tiles are laid in three layers or one layer depends upon the quality of tiles.
- Finally, three coats of plaster are applied for finishing the surface for draining rain water.

Since this type has found wide application in Madras state, it is called Madras terrace.

2. Reinforced cement concrete roof

This is the most widely adopted type for modern constructions. Simple slab roofs can be provided upto a span of 4.5 m and slabs supported by beam will be economical for higher spans. Concrete is spread over a wooden platform with steel rods placed over it. After curing the concrete for 7 to 14 days, when it attains strength the wooden supports and platform is removed. The total thickness of the slab will be usually 100 to 120 mm. A layer of brick jelly mixed with lime is spread on the top and is then finished with flooring tiles over a layer of cement mortar to get a leak proof surface.



Reinforced cement concrete roof

Pitched Roofs: In the areas of heavy rain fall and snow fall sloping roof are used. The slope of roof shall be more than 10° . They may have slopes as much as 45° to 60° also. The sloped roofs are known as pitched roofs. The sloping roofs are preferred in large spanned structures like workshops, factory buildings and ware houses. In all these roofs covering sheets like A.C. sheet, G.I. sheets, tiles, slates etc. are supported on suitable structures.

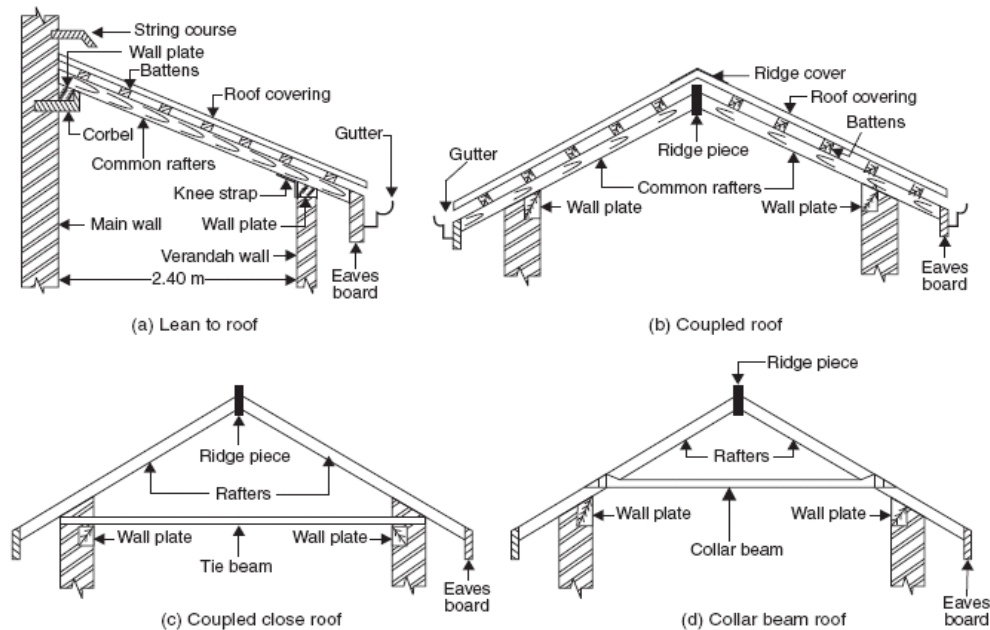
The pitched roofs are classified into

- (a) Single roofs (b) Double or purlin roofs (c) Trussed roofs.

(a) **Single Roof:** If the span of roof is less than 5 m the following types of single roofs are used.

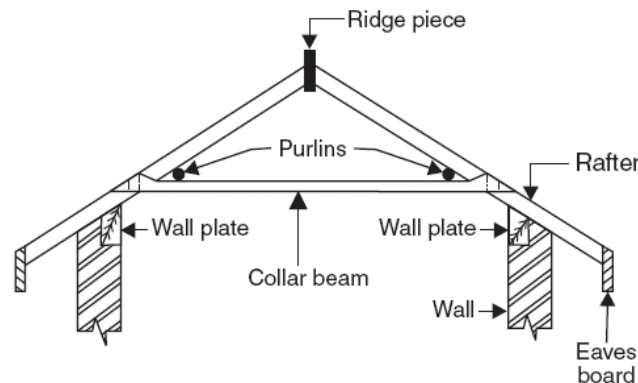
- (i) Lean to roofs (ii) Coupled roofs
(iii) Coupled-close roof (iv) Collar beam roof

In all these roofs rafters placed at 600 mm to 800 mm spacing are main members taking load of the roof. Battens run over the rafters to support tiles. Figure shows various types of single roofs.



Single roofs

(b) Double or Purlin Roofs: If span exceeds, the cost of rafters increase and single roof becomes uneconomical. For spans more than 5 m double purlin roofs are preferred. The intermediate support is given to rafters by purlins supported over collar beams. Figure 8.14 shows a typical double or purlin roof.



Double or purlins roofs

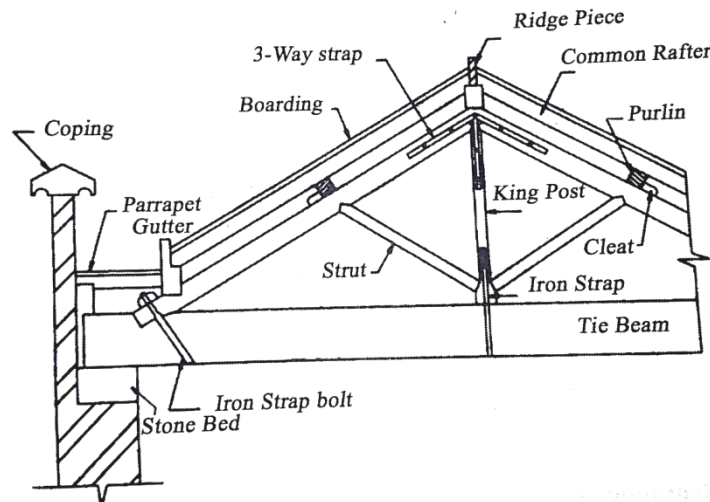
(c) Trussed Roof: When the span of the roof exceeds 5m and when there is no inside walls to support the purlins, framed structures known as trusses are provided at suitable interval along the length of the room.

The trussed roof system consists of three elements

- Rafters: to support roof coverings.

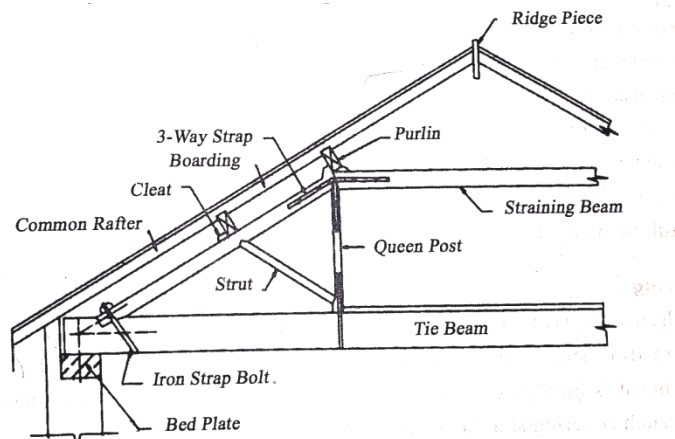
- Purlin: to provide intermediate supports to rafters.
- Trusses: to provide support to the ends of the purlins.

King Post truss



The vertical central post which connects the ridge piece with the tie beam is known as king post. The inclination compression members known as struts prevent the principle rafters from bending. This type of roof is suitable for spans varying from 5 to 9 m.

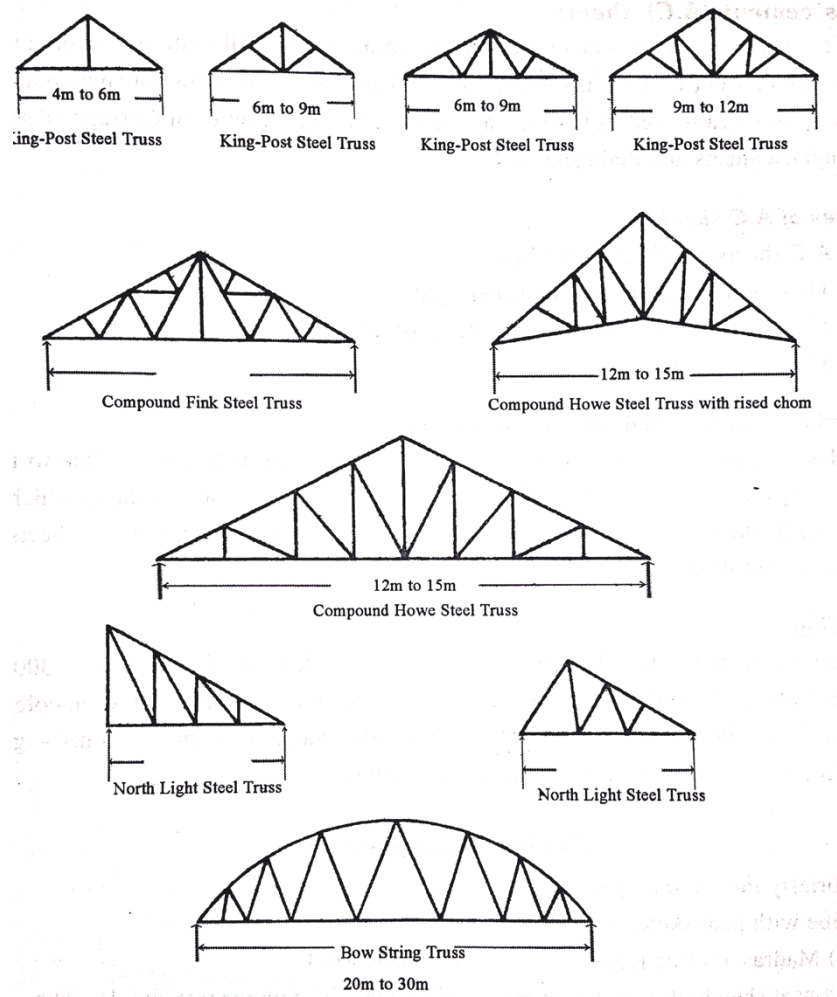
Queen post truss



This type of truss will have two vertical posts. The vertical posts are known as queen posts. The top of these posts are connected by a horizontal beam known as straining beam. Struts are provided to join the queen post to the principle rafter. The queen posts are tension members. Queen post trusses are suitable for spans from 9 to 14m.

Steel Trusses

When the span exceeds 10m, timber trusses become heavy and uneconomical, Mild steel is easily available in rolled sections of standard shapes and sizes. Most of the roof trusses are fabricated from angle sections because they can resist effectively tension as well as compression and their jointing is easy.



Types of steel truss

The roof truss is designed in such a way that the members carry only direct stresses and no bending stress is induced. The principle rafter as well as the main tie is generally made of two angle sections placed side by side, while the struts and ties are generally made of single angle sections. The members are joined together by using a gusset plate, either through rivets or by welding.

ROOF COVERING

A roof covering is a material which is placed over the frame work of pitched roof to protect the building from rain, sun, wind and other atmospheric agencies. The selection of various types of roofing materials depends upon the following factor

1. Type of building.
2. Type of roof frame work.
3. Initial cost.
4. Maintenance requirements.
5. Fabrication requirements.
6. Durability of the materials.
7. Climate of the locality.
8. Availability of the materials.

THATCH COVERING:

Thatch covering is the cheapest roof covering used in villages. Thatch covering consists of bundles of reeds or straw. The frame work to support consists of round bamboo rafters. It is very light, but it is highly combustible. It is unstable against high wind. The minimum thickness of thatch covering should be 15cm.

WOOD SHINGLES ROOFING:

Shingles are thin slab of wood. They are laid in a similar fashion as tiles and slates, this type of roofing is suitable for hilly area where local timber is easily available.

TILES:

This type of roofing is the oldest one and it is still preferred for residential buildings and country houses, because country tiles are manufactured from locally available earth. Plain or flat tiles are made of clay or concrete.

ASBESTOS CEMENT SHEET (A.C):

A.C sheets consist of cement and asbestos. Asbestos is a silky fibrous material found in veins in metamorphic rocks. It is found in several varieties by asbestos which is compound of magnesia and silica is used. A.C sheets are available in the varieties of corrugated sheets, semi-corrugated sheets and plain sheets.

Galvanized iron corrugated sheets (G.I):

G.I sheets are stronger than A.C sheets. These sheets are widely used. Due to their higher cost they are now replaced by A.C sheets. G.I sheets are made of iron sheets which are galvanized with zinc to protect them from rusting action of water. Therefore these sheets are also name as zinc sheets.

UNIT II -2 MARKS

- 1. State the types of floor commonly used. (Jan 2013) or How floors are classified based on floor finish?(Jan 2010)**

Mud or Murom flooring, cement flooring, marble flooring, mosaic flooring and tiled flooring

- 2. Define bearing capacity of soil and safe bearing capacity .(May 2013, May 2019) or Define safe bearing capacity.(Jan 2015,May 2016,Nov 2016)**

- The load carrying capacity of soils is called as their bearing capacity. In another words the property of soils to take the loads without failing is known as bearing capacity of soils.
- The maximum load the soil can carry safely without the risk of failure is known as its safe bearing capacity.

- 3. What are the functions of foundation?(Jan2010) or List the functions of the foundation.(Nov 2016 and 2017) or What are the crucial functions of foundations of buildings?(April/May 2014)**

- To distribute the load uniformly so that the load intensity below foundation is less than soil bearing capacity
- To provide a level base and support for superstructure
- To give stability against earthquake, wind, etc.
- Prevent cracks due to moisture, tilting and overturning

4. What are the structural components of a building?(April 2010)

Foundation, slab, lintel, column, beam are the structural components of a building which transfers load

5. State any two requirements of well planned roof.(Jan 2011)

- Durability against weather - rain, sun, wind cold and snow
- Water proofing and fire proofing
- Thermal insulation
- Sound insulation
- Strength and stability for carrying loads
- Well drained arrangement

6. What is simple footing?(April 2011)

It is the basic form of footing for column and wall. The member (column or wall) is provided with a base below it whose size is such that an offset is provided around the perimeter of the member. By doing so, the load on the member is spread to the soil below the footing.

7. What are the usual causes of failure of foundation?(April 2011)

- ✓ Increasing the bearing capacity of soil
- ✓ Distributing the load evenly on soil by suitable design and construction of foundation
- ✓ Designing the loads and components of structure like walls, columns and foundations such that it results in uniform or zero settlement and also negligible lateral pressure on superstructure
- ✓ Arresting the lateral movement of soil by constructing retaining walls, etc

8. State some methods to improve the bearing capacity of soil. [November, 2008]

- Increasing depth of foundation
- Draining the soil
- Cement grouting or chemical treatment
- Confining the soil or compacting the soil (by ramming moist soil, sand piles, rubble compaction, pre-loading, flooding the soil, vibrating the soil or Vibro-flotation method)

9. What are the components of substructure and superstructure of a building?[November, 2011]

Generally foundations are the part of substructure. All other components of buildings such as floors, walls, lintels, columns, beams, roofs, doors, windows, stairs, sunshades, etc. forms its superstructure.

10. What is roof trusses?(April 2012)

A frame work of slender members is used to support sloping roofs. These frames are known as trusses. Roof truss acts like a beam in a framed structure. Example: King post truss and Queen post truss.

11. What is a foundation (for building).(April/May 2016)

- It is part of sub-structure of a building, which is in direct contact with ground.
- It safely transmits and spreads the load from the superstructure to the soil below it.

12. Distinguish between sub structure and superstructure? Or What do you mean by superstructure? Give its function.(April/May 2014)

- Substructure is the lowest portion of a building which is usually considered to be located below the ground level. The substructure is nothing but the foundation which is provided to give stability to the building and to distribute the load coming from superstructure.
- Superstructure is that portion of a building which is constructed above the substructure. It is the major portion of a building which is used by human beings fully.

13. Define bonding in masonry.

[May, 2010]

Bond or bonding is the arranging of stones or bricks such that no vertical joint is formed. Thus no full surface of a stone or brick in a wall is in contact with full surface of the stone or brick which is above or below it. Lime or cement mortars are used to bond the building blocks, else they don't have sufficient strength and stability.

14. Differentiate load bearing and non load bearing wall.(Jan 2016)

- Masonry used for exterior walls, central main walls and some cross walls in a building are of load bearing walls. These walls should have adequate thickness such that it will sustain a slight eccentricity in loading.
- Partition loads are non-load bearing walls with strength, stability, sound insulation and fire resistance. This is an interior wall whose main function is to device the space within a building to rooms and other areas of varied use.

15. What are the types of deep foundation?(Nov 2016)

- Pile foundation
- Well foundation
- Caisson foundation

16. . Classify the roofs based on roof coverings.

[May, 2010]

Thatched roofs, asbestos cement sheet roofs, tiled roofs, timber or wooden roof, stone roof, RCC roof, steel roof, plastic or PVC roof, glass roof, etc.

17. What are the advantages and disadvantage of flat roof?

[May, 2010]

Advantage:

- It can be easily made fire proof
- It can be used for other activities like conducting some functions or storing materials

- It suits any type of building and vertical extension of building is easy
- Disadvantage:**
- Drainage is difficult and hence not suitable for rainy or snow clad areas
- It requires intermediate support for large spans and its dead weight or self weight is more
- Less aesthetic appearance

18. Differentiate between single and double roof? [JAN 2018]

- **Single roof:** It consists of only common rafters which are secured at the ridge and wall plate. These are used when span is less so that no intermediate support is required for the rafters. These are used up to a span of 5m. It is used for covering sheds and verandahs.
- **Double roof:** it is used up to a span of 3.5 m. It consists of two common rafters with ridge piece at the top and wall plates supporting them at the bottom. Horizontal onward forces will be acting on the wall plate and the wall below due to the tendency of the rafters to spread.

19. List any two types of foundation and indicate on which type of soil they are recommended. (Nov 2018)

Raft foundation –soft clay and peat soil

Under reamed pile foundation – Black cotton soil

UNIT II

DETAIL QUESTIONS:

1. What are the various types of soil classification and their use in building? Explain.(April/May 2014, Nov 2019)
2. Define bearing capacity of soils and give methods of improving the bearing capacity? (Nov 2017)
3. With a neat sketch explain the role and functions of various building components.(Nov 2016, April/May 2014, April/May 2016, Nov 2018)
4. What are the various causes of failures of foundation? What are the remedial measures. (Jan 2011, May 2019)
5. State the various functions of a foundation for a building?(May 2019)
6. With a sketch explain the various types of foundations used in building construction.(May 2013, May 2016)
7. a) Draw a neat sectional elevation of a building and show its basic components.(Jan 2016)
b) Explain the types of shallow foundation with neat sketch.(Jan 2016)
8. Compare stone masonry and brick masonry. (Jan 2010, April 2010, Jan, May 2013)

9. What are the points to be considering while constructing brick masonry.(Jan 2011)
10. List the types of stone masonry. Explain types of rubble masonry with neat sketch.(April/May 2014, April 2012)
11. a) Differentiate English and Flemish bond with sketches.(Jan 2016)
b) Explain different types of roofing. Also state their merits and demerits (Jan 2016, nov 2018).
12. Explain the different types of bonds used in brick masonry.(May 2016)
13. Explain the construction of granolithic flooring, where do you recommend. (Jan 2013)
14. Explain the various types of flooring used for residential buildings.(Jan 2015, April/May 2016, Nov 2018)
15. Classify types of flooring. Discuss the cement concrete flooring and marble flooring with neat sketch.(May 2019)
16. Explain the various factors influencing the selection of flooring materials.(April/May 2014)
17. What are roofs? Explain the salient features of reinforced cement concrete roofs.(Nov 2016)